上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

SJTU

基于声传播的智能定位系统（F题）赛题解析和分享

韩韬

2023年10月14日

饮 水 思 源 • 爱 国 荣 校

主要内容

1. 器材选型与制备

2. 基本要求1解析

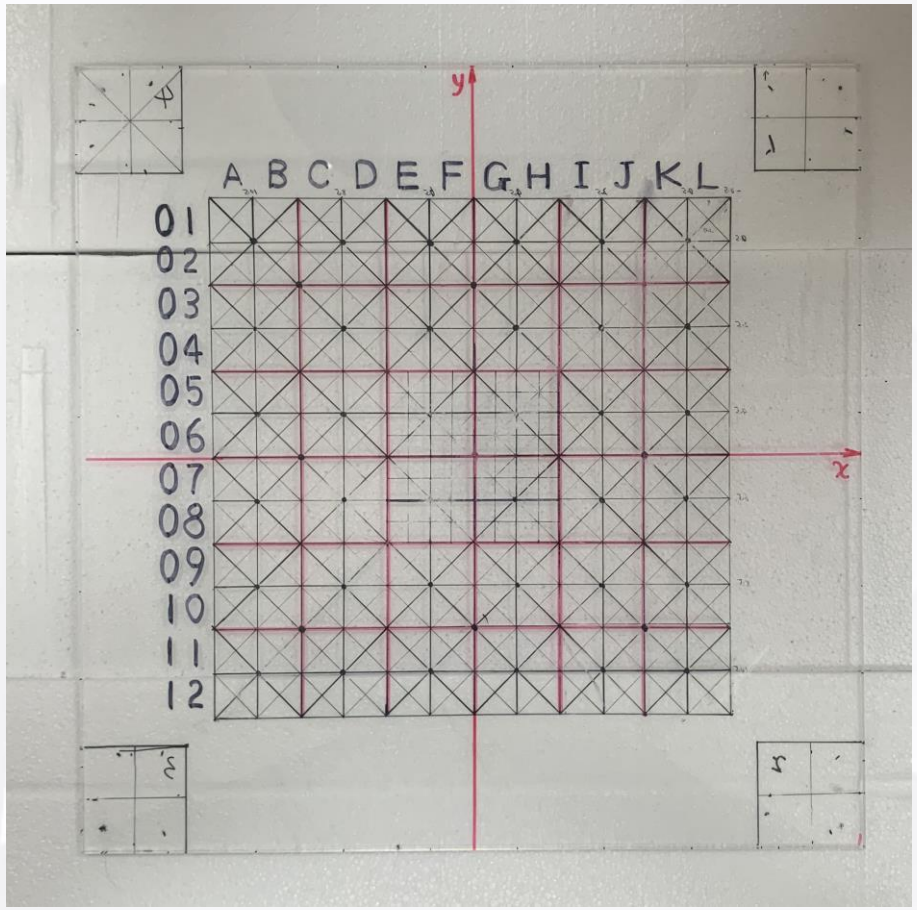
3. 基本要求2解析

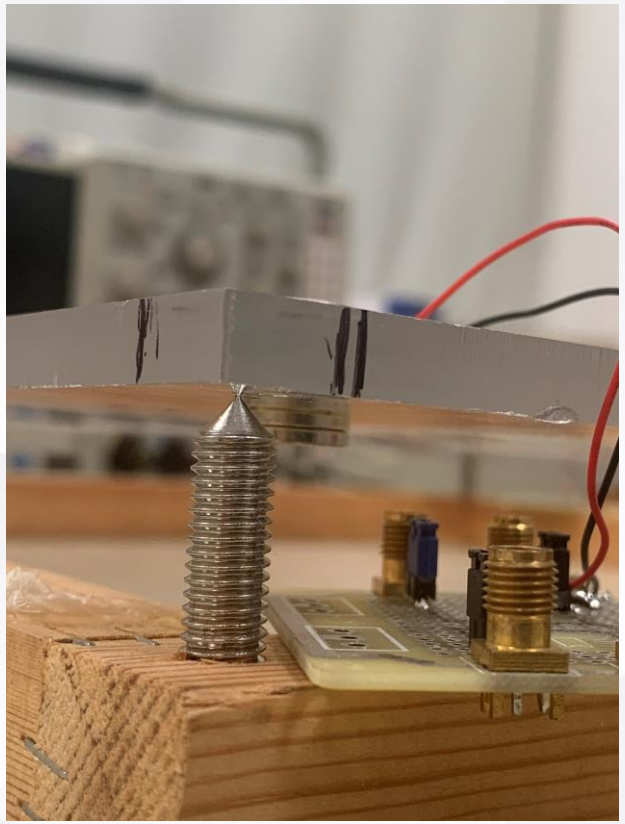
4. 发挥部分解析

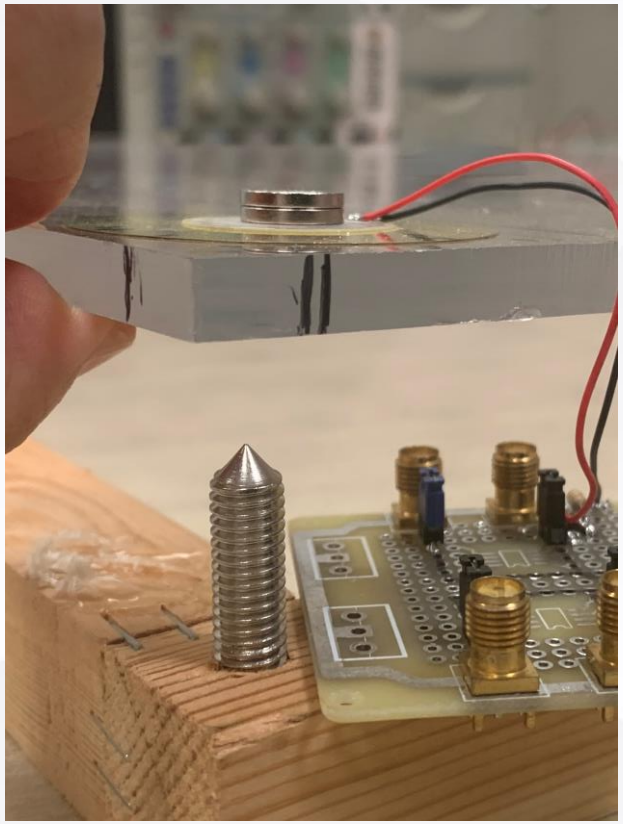
5. 总体技术方案实例

1. 器材选型与制备

◆ 正方形亚克力材质平面板





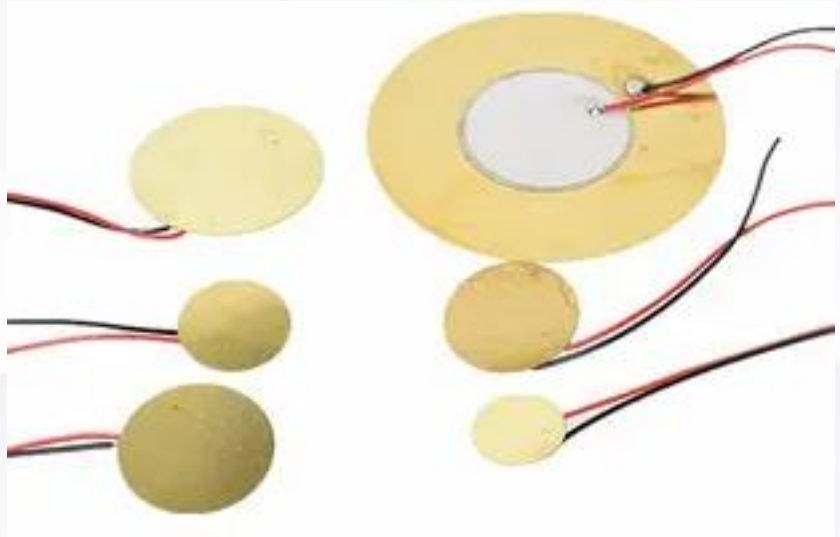


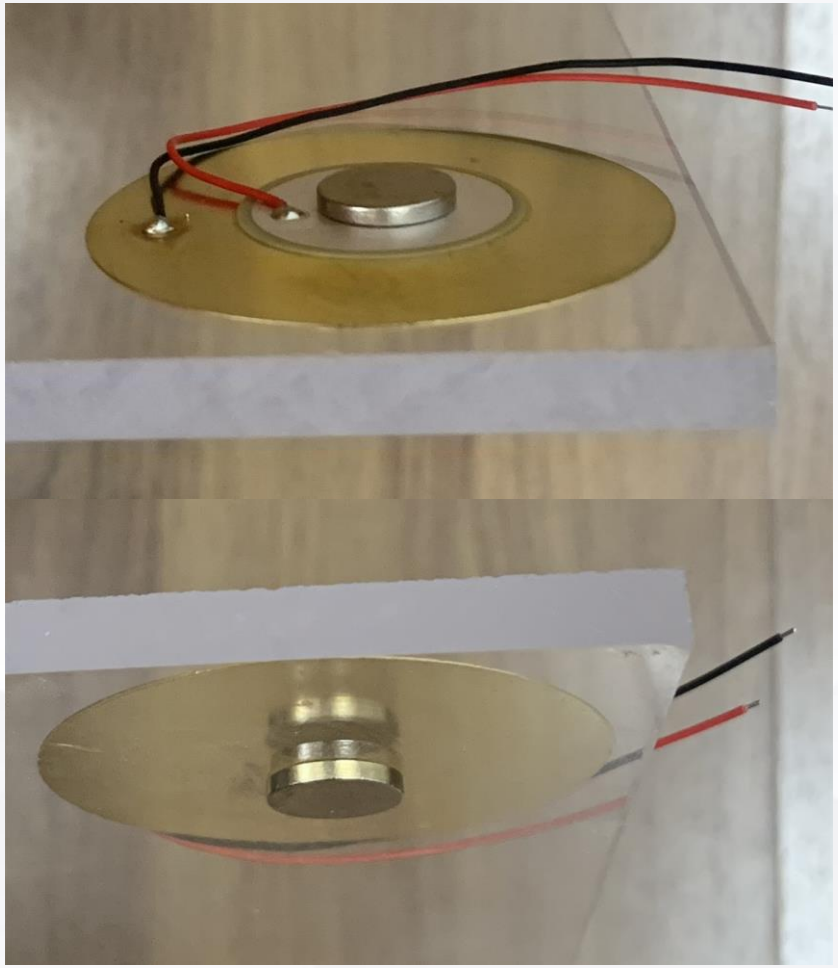
四个角尖端支撑

◆ 电声转换、声电转换器件

扬声器、蜂鸣器、压电陶瓷蜂鸣片

- ◆ 频率响应宽
- ◆ 电压驱动
- ◆ 与亚克力板易于紧贴（无需耦合剂）





- ◆ 不使用胶水将蜂鸣片与亚克力板黏合
- ◆ 做好标记，易于快速恢复

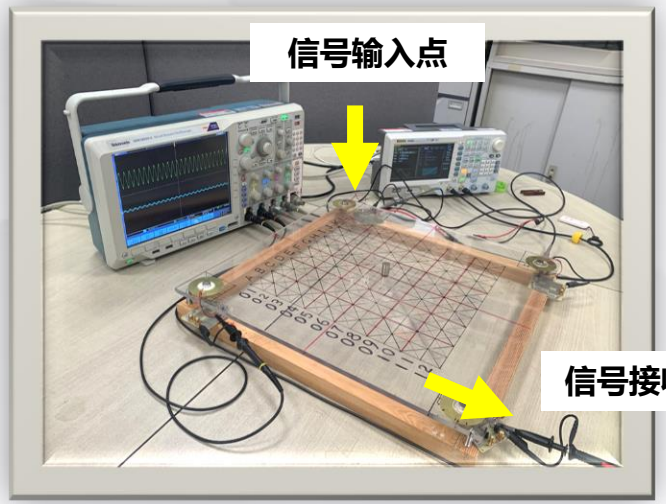
◆ 足够小端面的木锤
（用于敲击亚克力板）

◆ 钕铁硼磁铁圆片

2. 基本要求1：周期扫频信号发生、传导检测

频率范围15kHz~20kHz，周期5秒

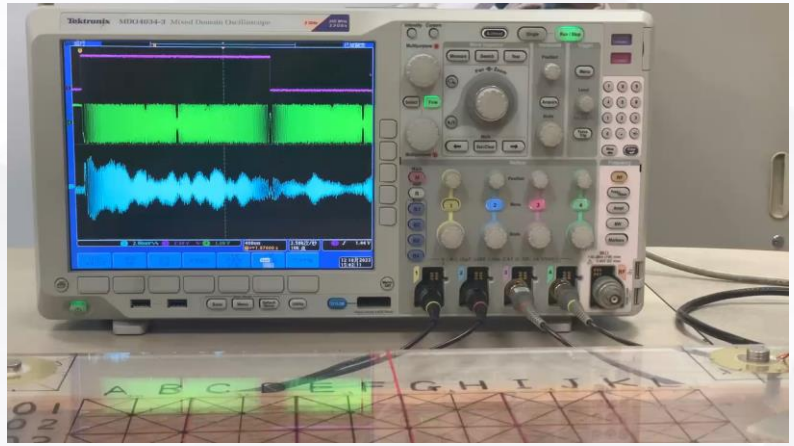
- 解题提示、保证基础得分
- 测试时减小噪音

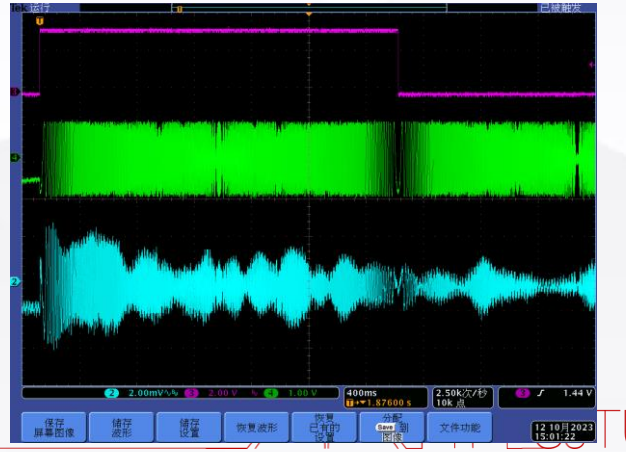


扫频信号发生 → 电声转换 → 声波传导 → 声电转换 → 接收端前置放大 → 示波器观测

交流电压信号幅度：
建议约 1Vpp

前置放大器增益：
建议50~100倍

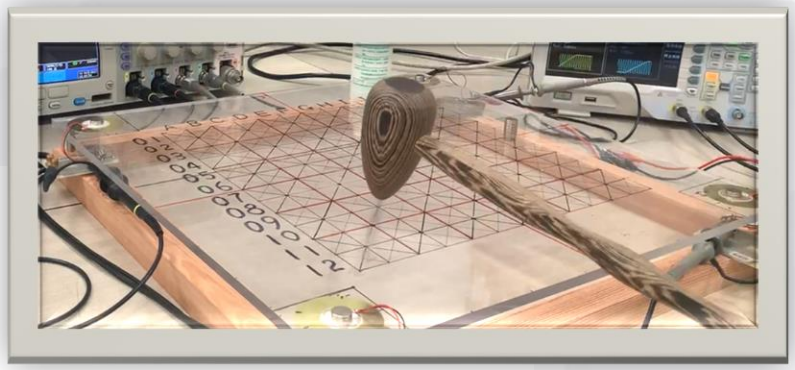


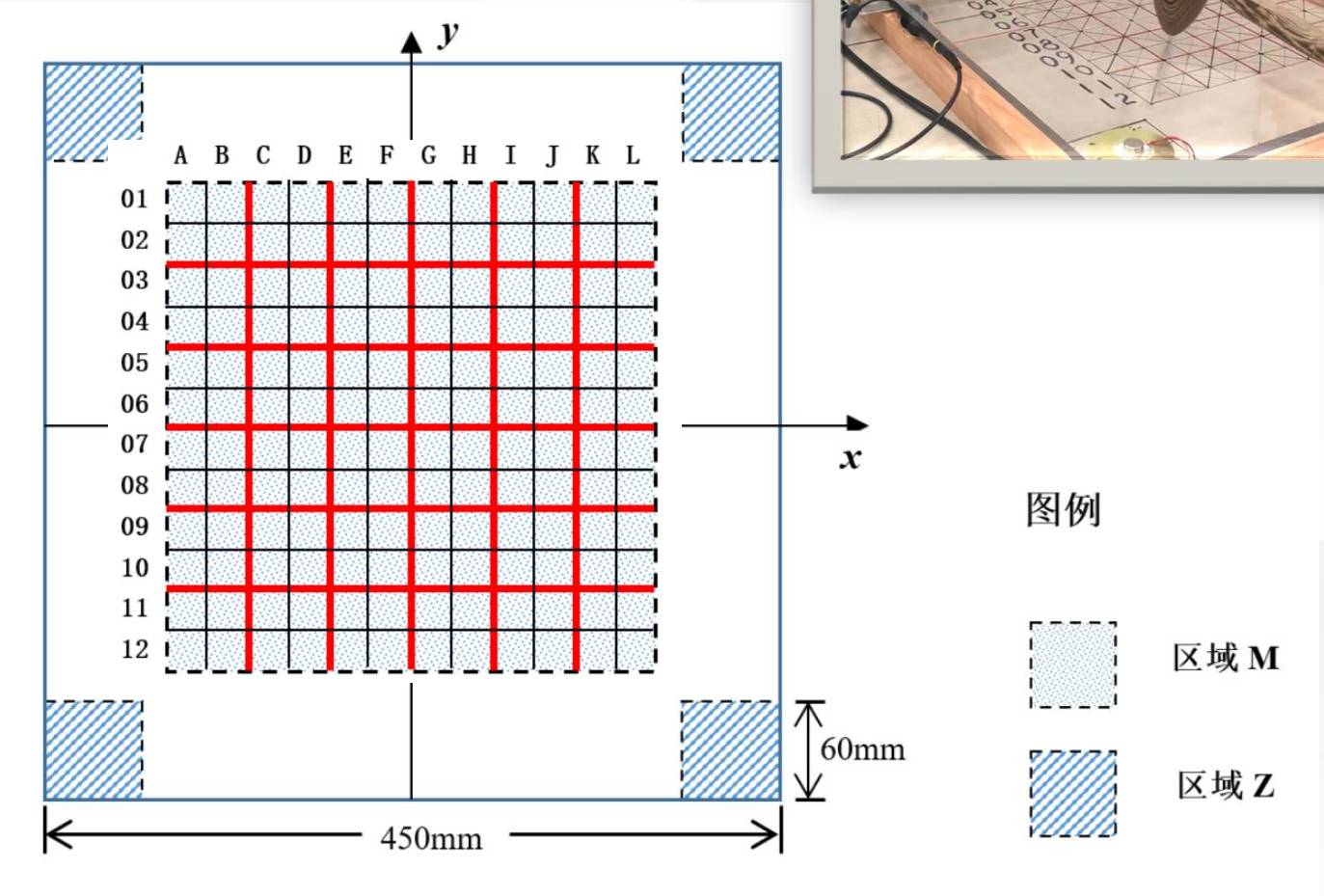


扫频演示视频

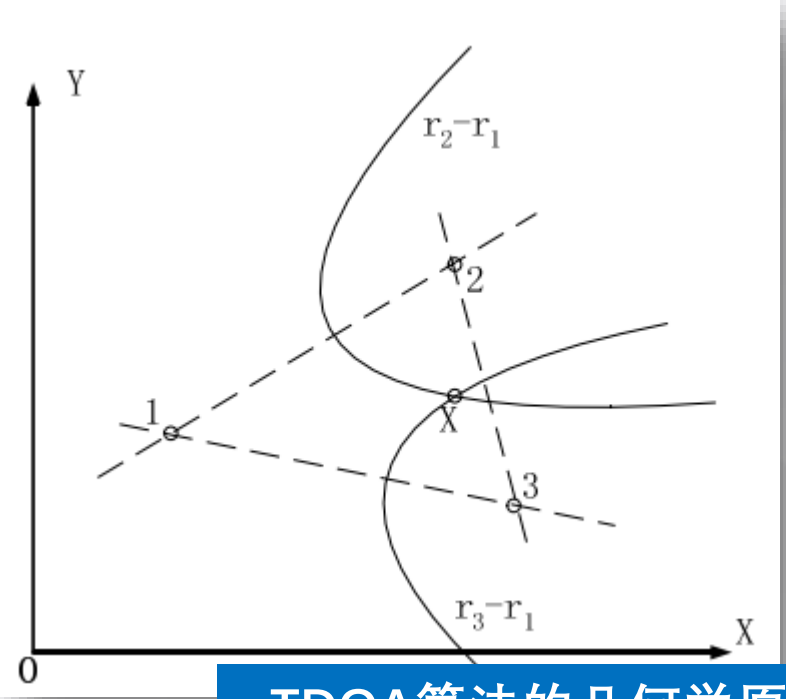
3. 基本要求2：敲击定位

◆ 常用定位原理





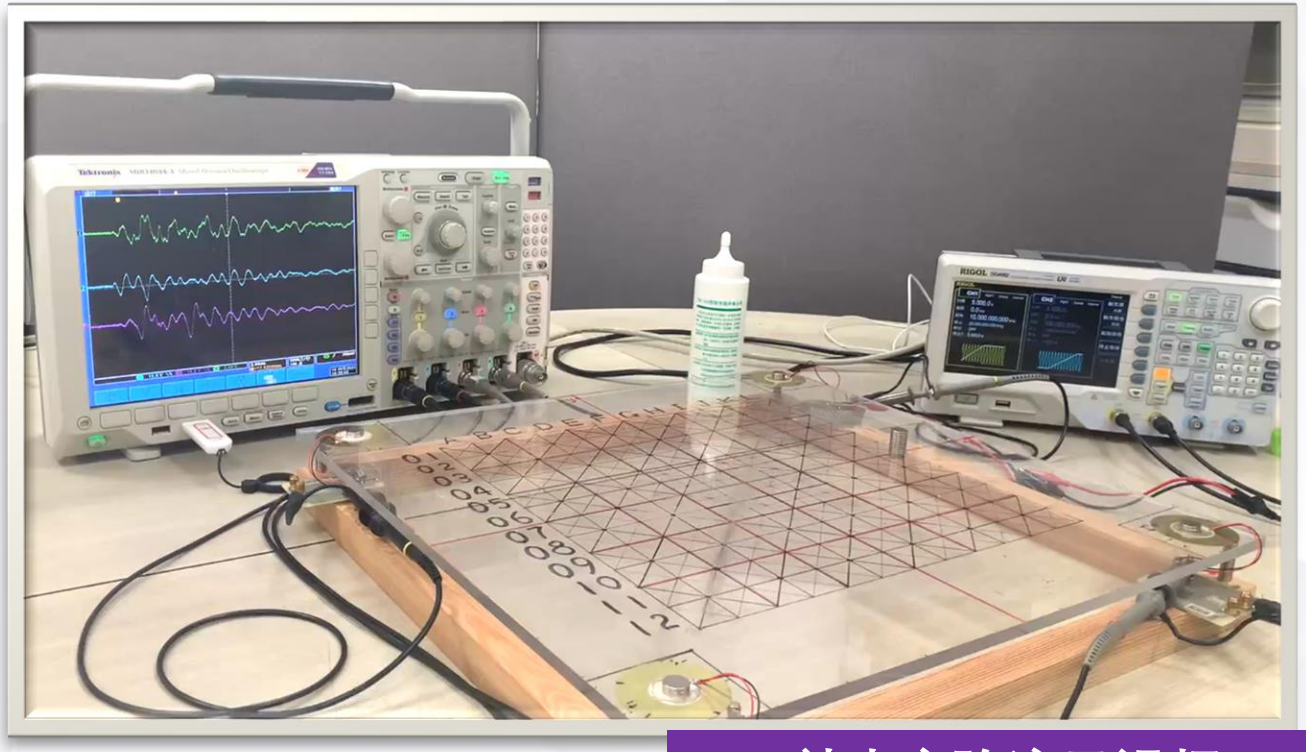
信号到达时间差
TDOA（Time Difference of Arrival）



TDOA算法的几何学原理

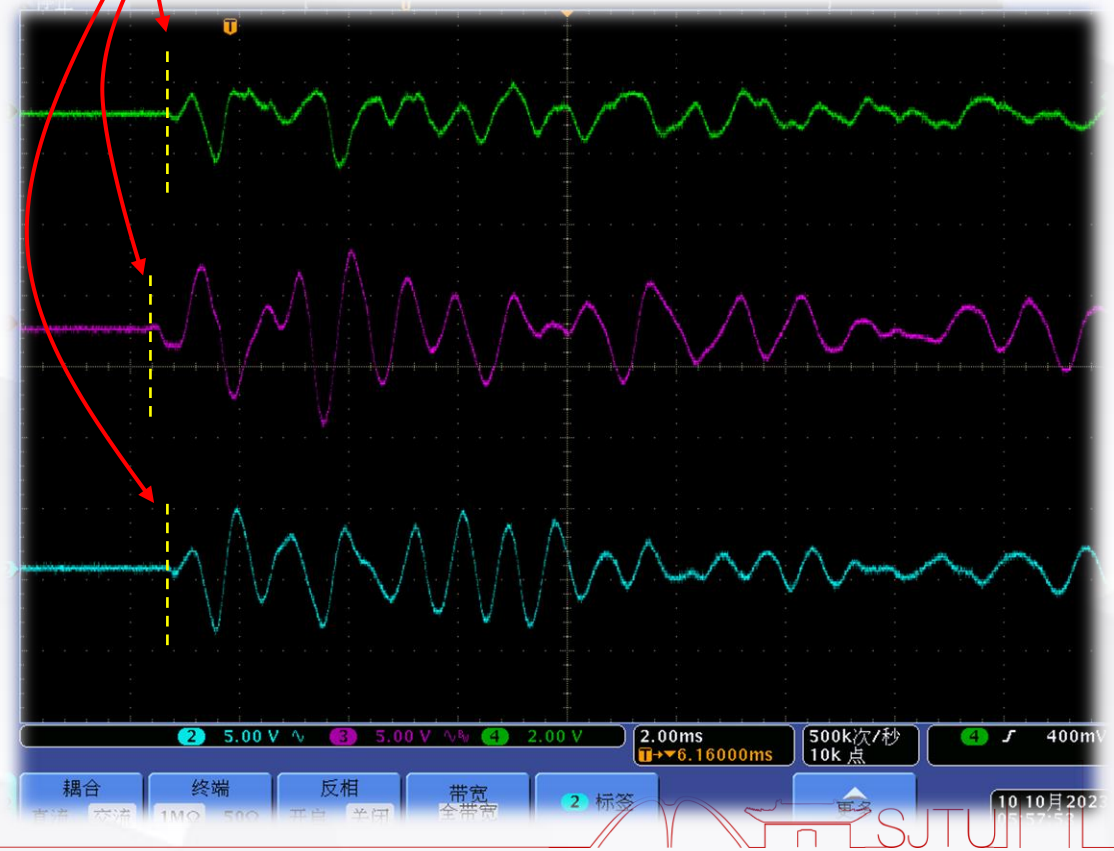
基本要求2

◆ 实际信号与定位方法



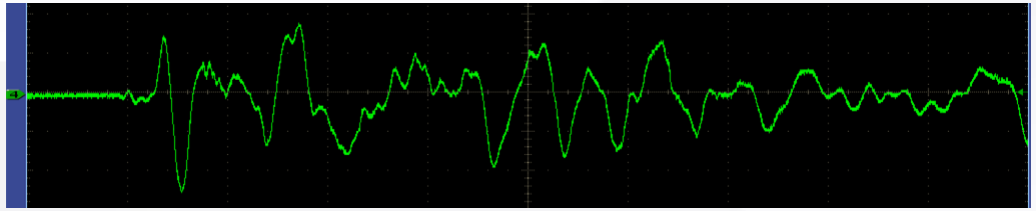
敲击实验演示视频

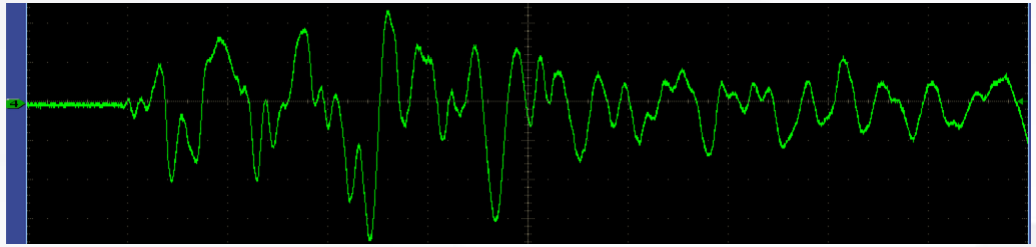
信号各自的起始位置

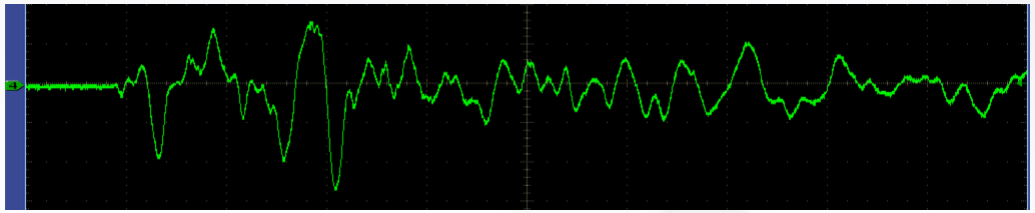


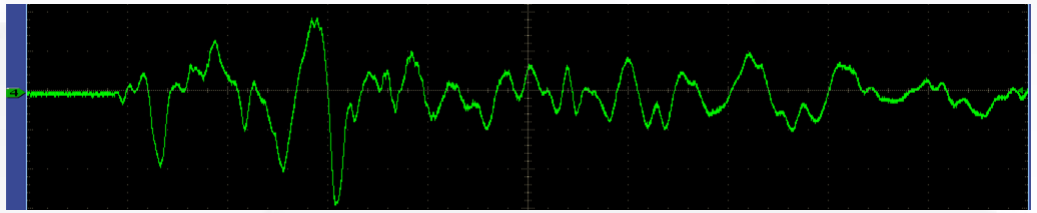
基本要求2

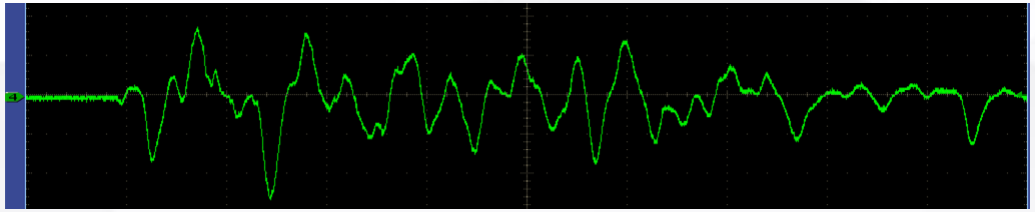
◆ 波形相关计算 + 识别判决定位





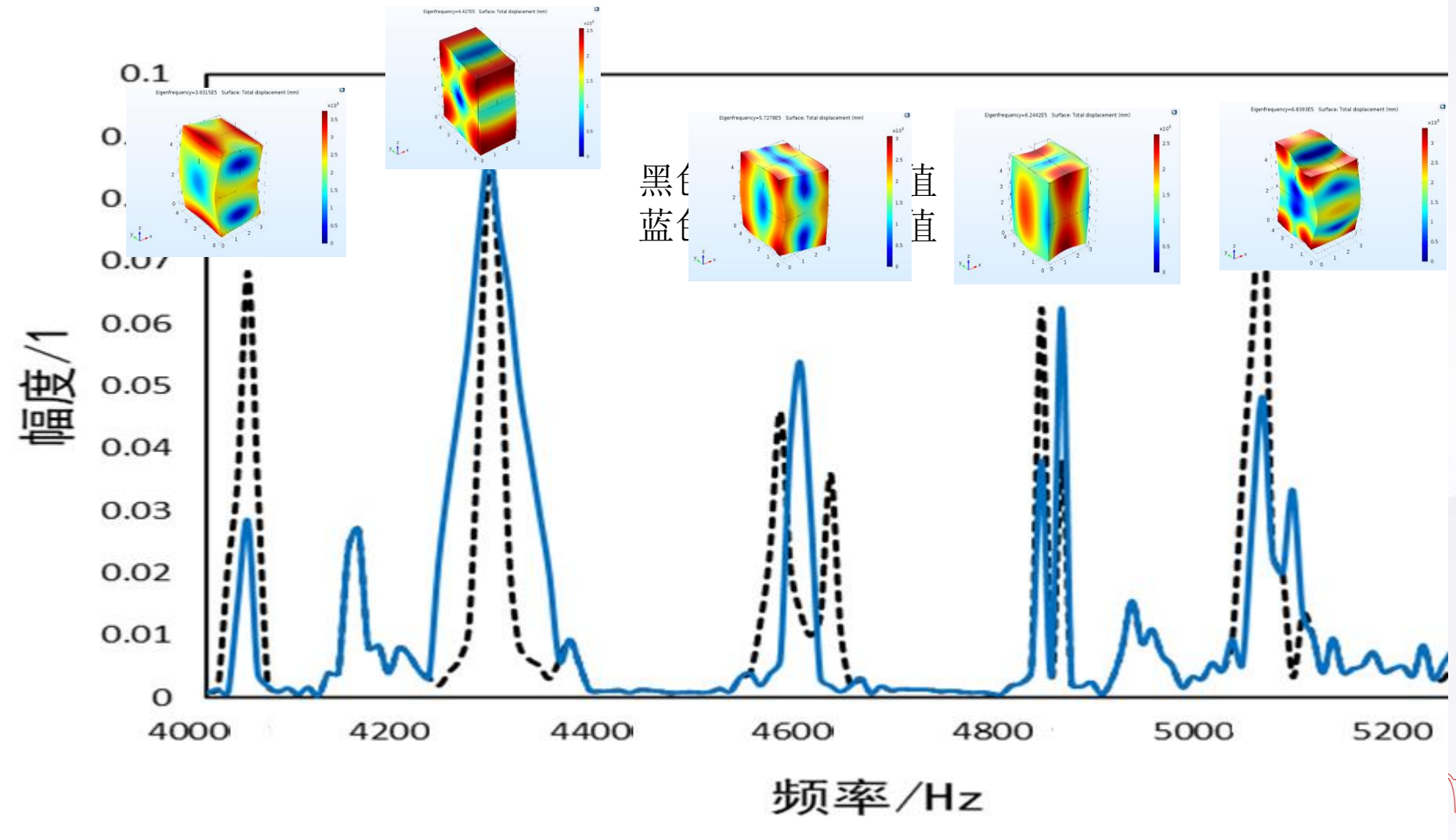






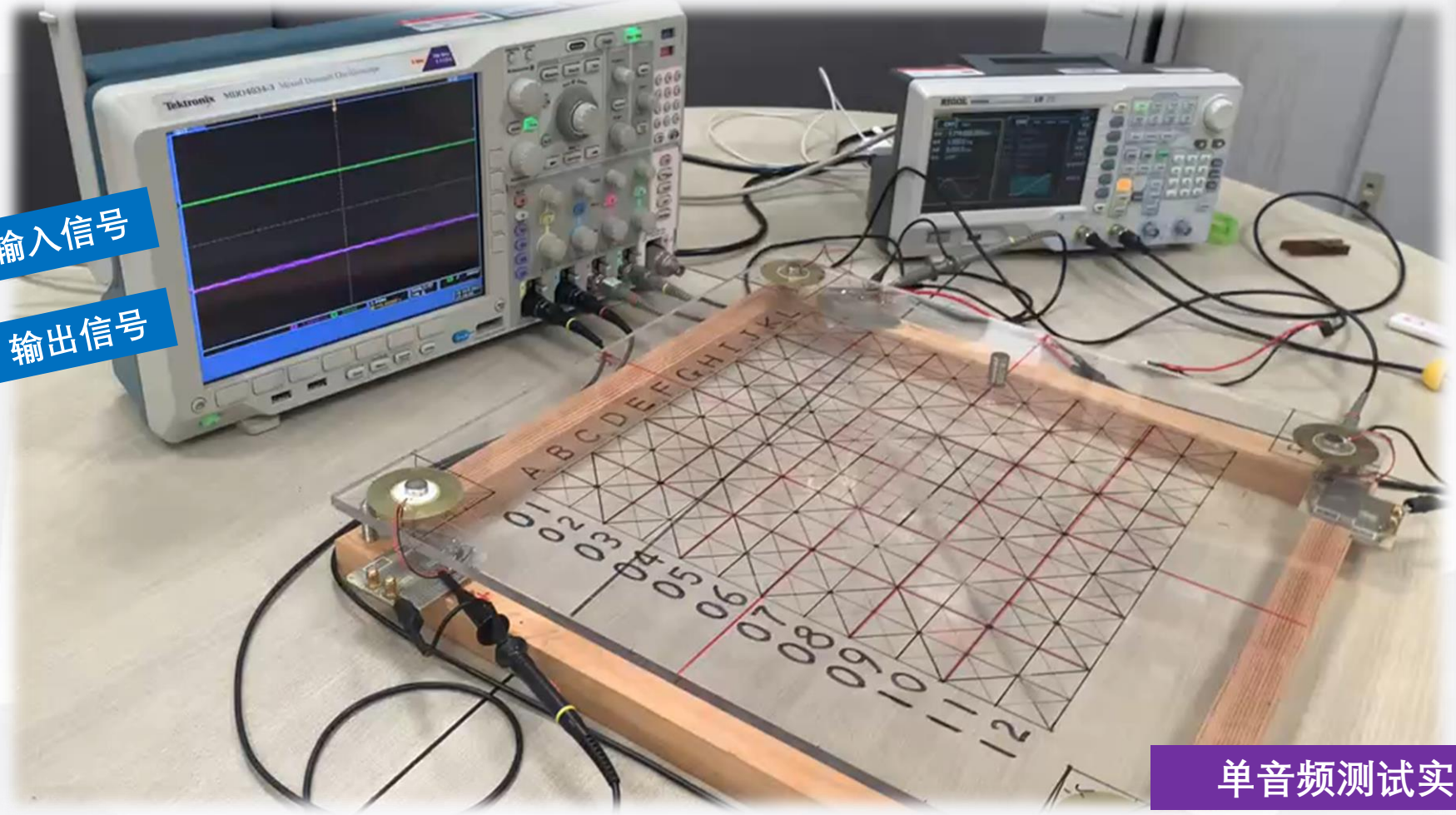
4. 发挥部分解析

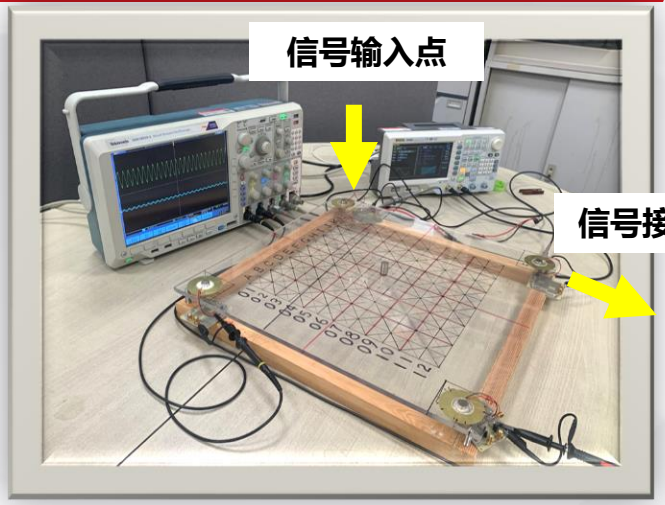
平板上某位置放置物件，如何确定它所在位置?



发挥部分解析

◆ 单音频测试，验证物件位置与频率特性的关系



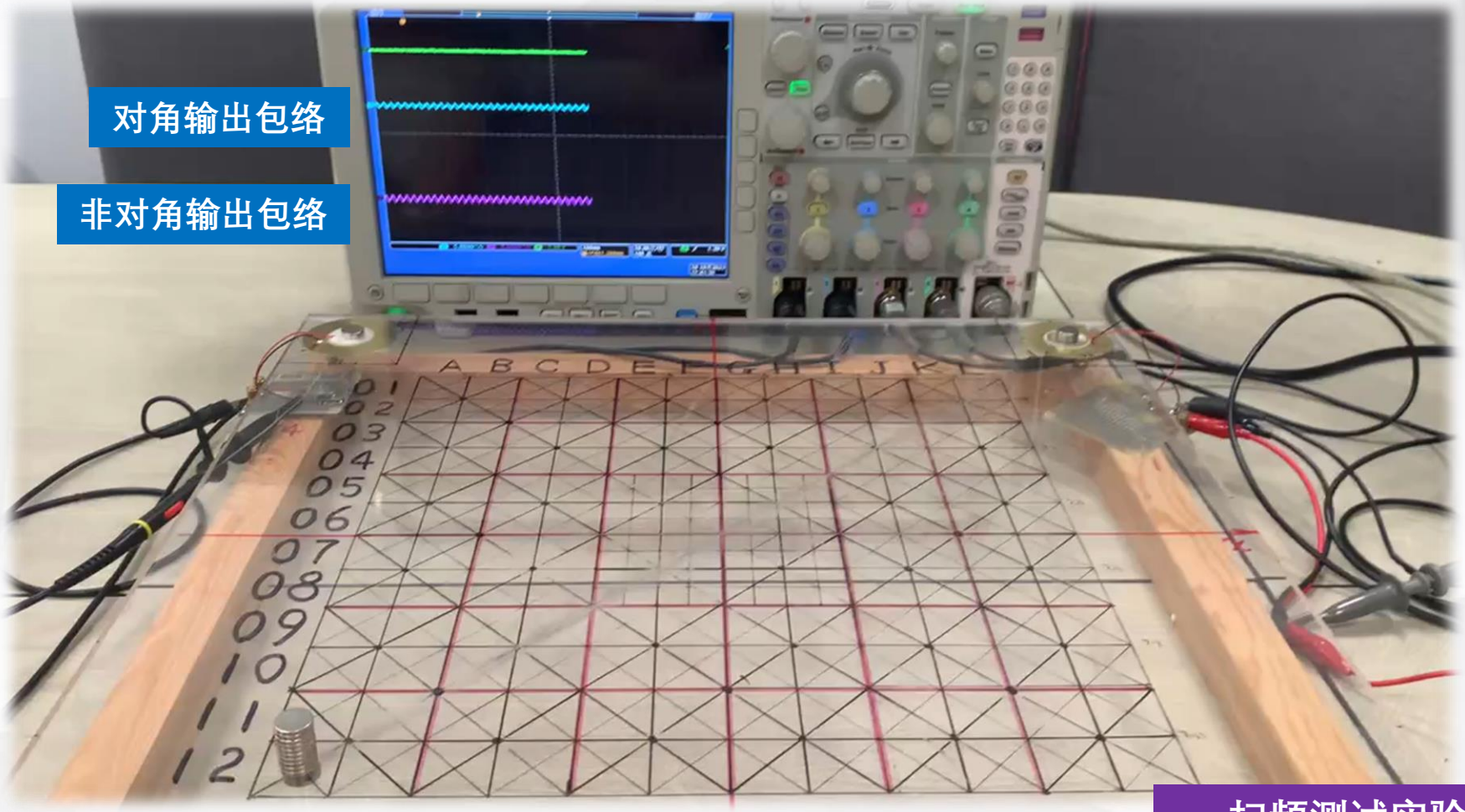


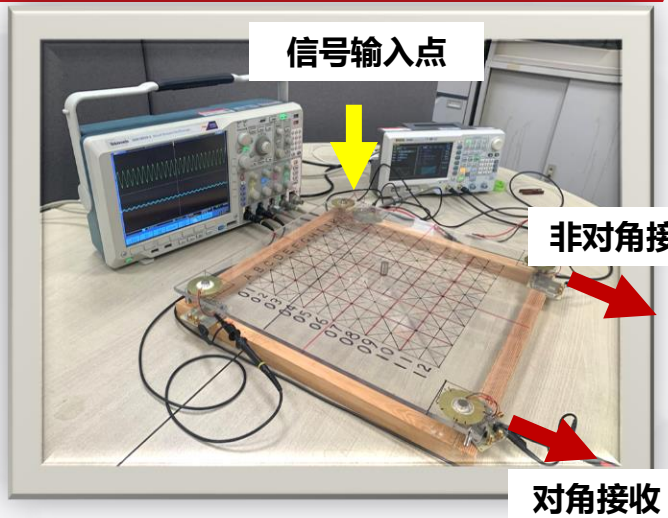
单音频正弦波频率约1700Hz

单音频测试实验视频

发挥部分解析

◆ 扫频测试



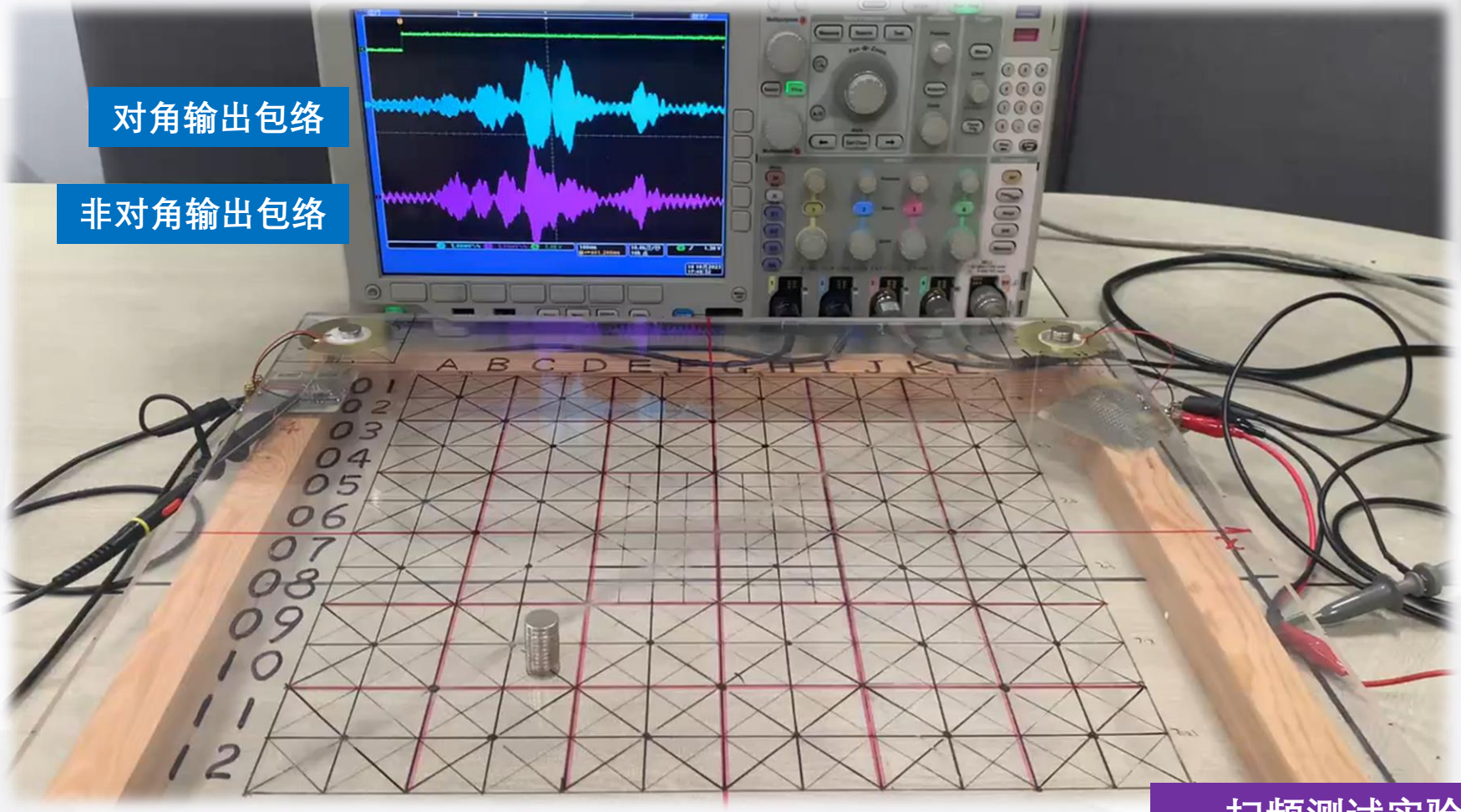


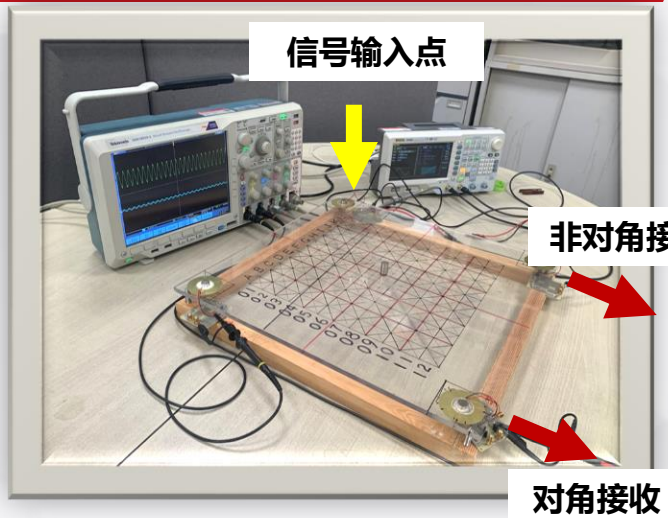
起始频率：500Hz
终止频率：5000Hz
扫描时间：2.5秒
停止间隔：2秒

扫频测试实验视频1

发挥部分解析

◆ 扫频测试





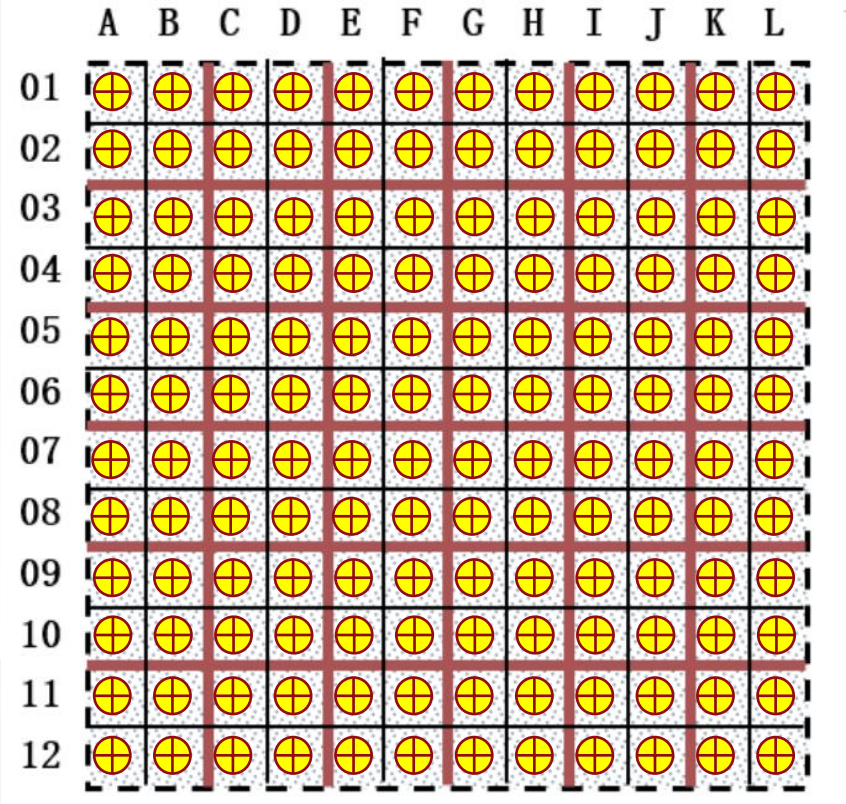
起始频率：500Hz
终止频率：5000Hz
扫描时间：2.5秒
停止间隔：2秒

扫频测试实验视频2

发挥部分解析

◆ 标定

将物件轮流摆放在每个格子，分别测量和记录频率特性



相关度计算 + 匹配识别
（基于12x12 共144套数据档案）

物件在(A,01)时的频率特性

物件在(A,02)时的频率特性

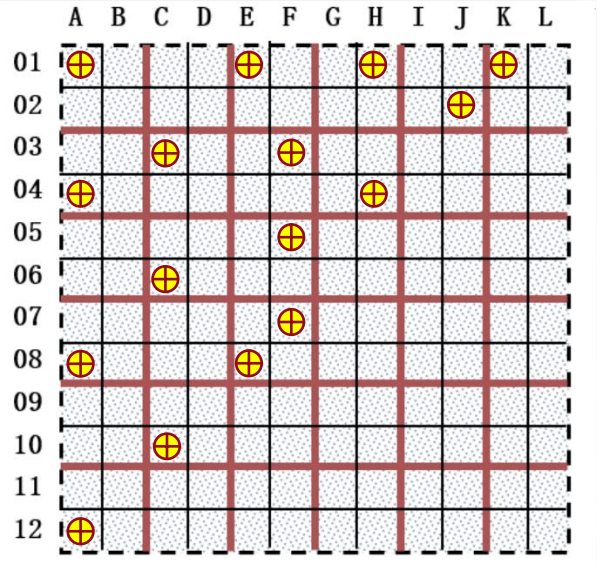
⋮

物件在(L,12)时的频率特性

发挥部分的解析

◆ 系统再次标定

仅挑选少部分格子，
重新测试和标定

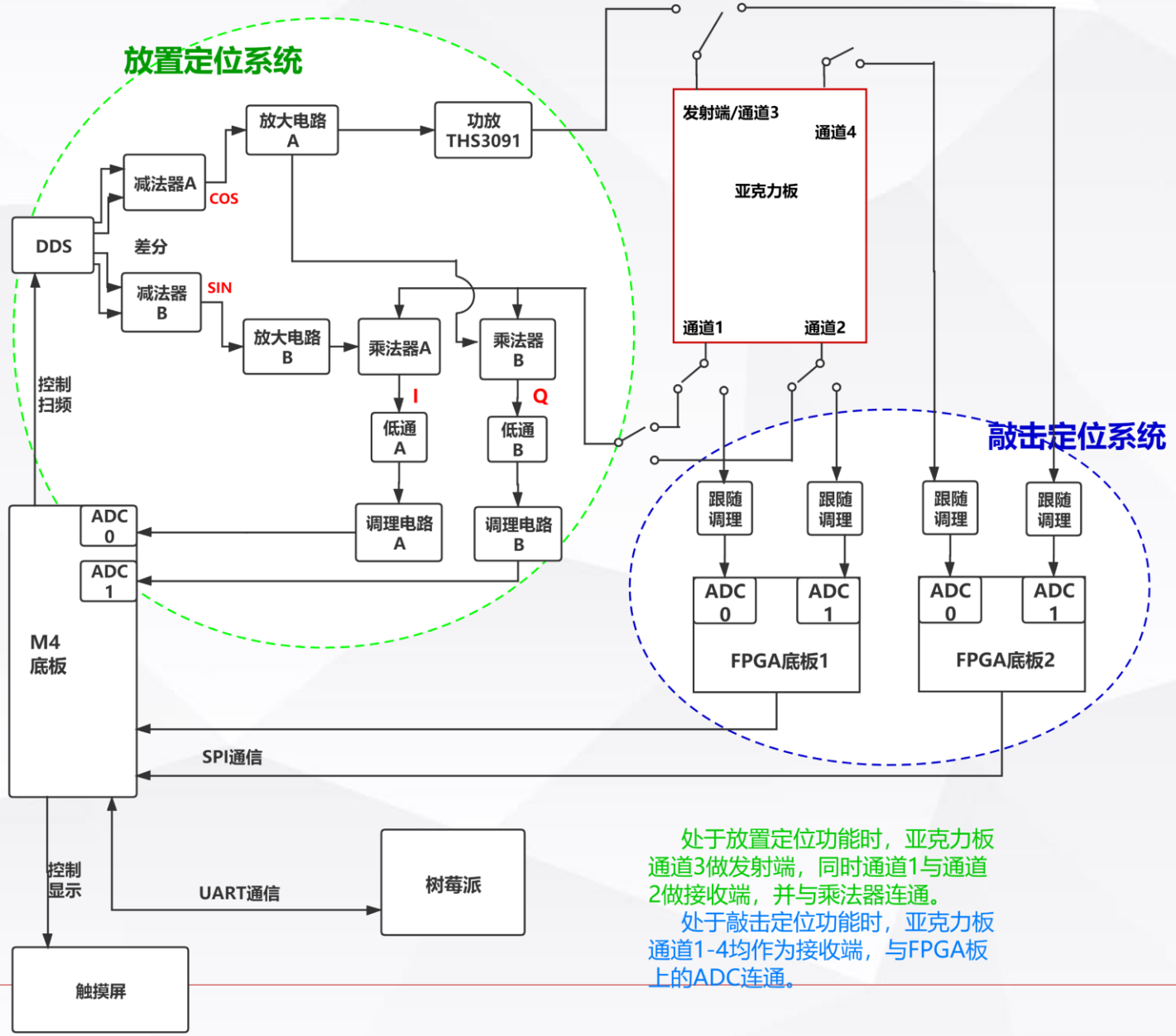


拟合插值等传统算法
或
神经网络机器学习算法

频率特性数据档案

- 物件在(A,01)时的频率特性
- 物件在(A,02)时的频率特性
- ⋮
- 物件在(L,12)时的频率特性
- 物件在任意位置的频率特性

5. 总体方案实例（系统组成框图）



处于放置定位功能时，亚克力板通道3做发射端，同时通道1与通道2做接收端，并与乘法器连通。

处于敲击定位功能时，亚克力板通道1-4均作为接收端，与FPGA板上的ADC连通。

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感谢各位

饮 水 思 源　爱 国 荣 校